

Appendix A. Station-wise sensitivity analysis of probabilistic inferences

This appendix presents the additional station-wise sensitivity analysis performed. The inference scenarios reported in Figures 8, 9 and 10 of the manuscript were reproduced separately for Forte de Copacabana, Jacarepaguá, Marambaia and Vila Militar.

The objective of this analysis was not to train independent station-specific Bayesian Networks, but to verify whether the conditional probability distributions obtained from the fitted model showed similar qualitative behavior across the four monitoring locations. The results indicate that the main physical patterns are preserved across stations, although the magnitude of the estimated probabilities varies in some scenarios. This suggests that local factors may modulate the strength of the relationships among meteorological variables and rainfall occurrence.

Cases with insufficient observations for a reliable station-specific estimate were not interpreted and are indicated as *Insufficient data*.

Table 1: Station-wise conditional probabilities for the humidity scenario corresponding to Figure 8.

Rainfall state	Humidity level	Copacabana	Jacarepaguá	Marambaia	Vila Militar
No Rain	Low	98.66%	98.63%	99.63%	98.83%
	Moderate	97.42%	96.54%	98.65%	96.04%
	High	86.59%	86.75%	87.86%	86.48%
Rain	Low	1.32%	1.37%	0.37%	1.15%
	Moderate	2.58%	3.43%	1.35%	3.93%
	High	13.36%	13.16%	12.09%	13.44%
Extreme Rain	Low	0.02%	0.00%	0.01%	0.02%
	Moderate	0.00%	0.03%	0.01%	0.02%
	High	0.05%	0.09%	0.05%	0.08%

Table 2: Station-wise conditional probabilities for the global radiation scenario corresponding to Figure 8.

Rainfall state	Radiation level	Copacabana	Jacarepaguá	Marambaia	Vila Militar
No Rain	Low	85.82%	82.01%	86.53%	89.38%
	Moderate	96.35%	96.78%	98.03%	96.10%
	High	98.33%	98.14%	99.62%	98.46%
Rain	Low	14.12%	17.86%	13.44%	10.55%
	Moderate	3.65%	3.19%	1.96%	3.88%
	High	1.66%	1.85%	0.38%	1.53%
Extreme Rain	Low	0.06%	0.13%	0.04%	0.07%
	Moderate	0.00%	0.03%	0.00%	0.02%
	High	0.01%	0.01%	0.00%	0.01%

Table 3: Station-wise conditional probabilities for the wind-speed scenario corresponding to Figure 8.

Rainfall state	Wind-speed level	Copacabana	Jacarepaguá	Marambaia	Vila Militar
No Rain	Low	93.21%	90.99%	92.70%	92.47%
	Moderate	85.35%	<i>Insufficient data</i>	89.50%	86.45%
	High	92.17%	92.76%	93.67%	92.47%
Rain	Low	6.76%	8.96%	7.28%	7.50%
	Moderate	14.55%	<i>Insufficient data</i>	10.43%	13.39%
	High	7.80%	7.20%	6.30%	7.44%
Extreme Rain	Low	0.03%	0.06%	0.02%	0.03%
	Moderate	0.10%	<i>Insufficient data</i>	0.07%	0.15%
	High	0.03%	0.05%	0.03%	0.09%

Note. For Jacarepaguá, the moderate wind-speed condition contained too few observations to support a reliable station-specific estimate. Therefore, the corresponding probabilities were not interpreted.

Table 4: Station-wise conditional probabilities for the moderate wind-speed and humidity scenario corresponding to Figure 9.

Rainfall state	Humidity level	Copacabana	Jacarepaguá	Marambaia	Vila Militar
No Rain	Low	99.02%	<i>Insufficient data</i>	99.73%	98.32%
	Moderate	96.26%	<i>Insufficient data</i>	98.11%	88.86%
	High	76.75%	<i>Insufficient data</i>	74.35%	58.78%
Rain	Low	0.98%	<i>Insufficient data</i>	0.26%	1.68%
	Moderate	3.74%	<i>Insufficient data</i>	1.87%	11.12%
	High	23.11%	<i>Insufficient data</i>	25.51%	39.78%
Extreme Rain	Low	0.00%	<i>Insufficient data</i>	0.01%	0.00%
	Moderate	0.00%	<i>Insufficient data</i>	0.02%	0.01%
	High	0.14%	<i>Insufficient data</i>	0.14%	1.43%

Note. This scenario conditions the inference on moderate wind speed. For Jacarepaguá, the number of observations under this evidence configuration was insufficient for a reliable station-specific interpretation.

Table 5: Station-wise conditional probabilities for the low global radiation and humidity scenario corresponding to Figure 10.

Rainfall state	Humidity level	Copacabana	Jacarepaguá	Marambaia	Vila Militar
No Rain	Low	98.08%	97.51%	98.86%	98.60%
	Moderate	96.62%	95.34%	97.39%	95.62%
	High	80.48%	76.35%	81.32%	83.81%
Rain	Low	1.92%	2.49%	1.06%	1.39%
	Moderate	3.38%	4.66%	2.59%	4.35%
	High	19.45%	23.48%	18.62%	16.09%
Extreme Rain	Low	0.00%	0.00%	0.08%	0.00%
	Moderate	0.00%	0.00%	0.02%	0.03%
	High	0.07%	0.17%	0.06%	0.10%